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METHOD AND APPARATUS FOR LASER DRILLING BLIND VIAS

Abstract

In an embodiment, a method of forming a blind via in a substrate comprising a mask layer, a conductive layer, and a dielectric layer is provided. The method includes detecting the mask layer by a sensor, the mask layer providing a substrate surface; determining a property of the blind via, the property comprising one or more of a top diameter, a bottom diameter, a volume, or a taper angle; focusing a Gaussian laser beam, under laser process parameters, at the substrate surface to remove at least a portion of the mask layer; adjusting the laser process parameters based on the property; and focusing the laser beam, under the adjusted laser process parameters, to remove at least a portion of the dielectric layer within the volume to form the blind via. The mask layer can be pre-etched. Apparatus for forming a blind via in a substrate are also provided.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation application of co-pending application Ser. No. 17/508,512, filed Oct. 22, 2021 which is a continuation of Ser. No. 16/928,252, entitled “Method and Apparatus for Laser Drilling Blind Vias” and filed Jul. 14, 2020. Both cases are herein incorporated by reference in their entireties.

FIELD

[0002] Embodiments of the present disclosure generally relate to methods and apparatus for drilling blind vias.

DESCRIPTION OF THE RELATED ART

[0003] With increasing demand for miniaturized electronic devices and components, the demand for faster processing capabilities with greater circuit densities imposes demands on the materials, structures, and processes used in the fabrication of integrated circuit chips and printed circuit boards. Incorporating blind vias enable more connections and greater circuit density. Laser drilling is an established method for forming blind vias.

[0004] Alongside the trend toward smaller vias, e.g., diameters from 5 μm to 10 μm , for higher interconnect density, certain blind via quality parameters remain. These include via top diameter and roundness, via bottom diameter and roundness, via taper angle (80° or more), and via pad cleanliness. However, conventional laser-drilling technologies cannot form quality blind vias having such small diameters for high-volume manufacturing.

[0005] Conventional approaches for laser drilling blind vias use nanosecond-pulsed 355 nm ultraviolet (UV) lasers or pulsed CO₂ lasers to directly drill blind vias in panels. However, both these approaches require complex beam shaping optics to convert the Gaussian laser beam profile emitted by the laser sources into a top-hat shaped beam profile. Further, top-hat shaped laser beam profiles generally have a very short depth-of-field within which the laser beam profile remains equal in intensity. Moreover such top-hat beam profiles are not capable of drilling diameters less than 40 μm , particularly 5 μm to 10 μm , in a consistent and cost-effective manner. Moreover, conventional laser drilling methods cannot enable blind vias having a taper angle of 80° or more.

[0006] There is a need for new and improved methods and apparatus for drilling high-quality, small-diameter blind vias in high volumes.

SUMMARY

[0007] Embodiments of the present disclosure generally relate to methods and apparatus for drilling blind vias.

[0008] In an embodiment, a method of forming a blind via in a substrate is provided. The method includes conveying the substrate to a scanning chamber of a laser drilling system, the substrate comprising a conductive layer, a dielectric layer disposed on at least a portion of the conductive layer, and a mask layer disposed on at least a portion of the dielectric layer. The method further includes detecting the mask layer by a sensor of the laser drilling system, the mask layer providing a substrate surface, and determining one or more properties of the blind via, the one or more properties comprising: a top diameter of the blind via and a bottom diameter of the blind via, the

blind via having a height from the top diameter to bottom diameter, the top diameter being greater than the bottom diameter; a volume of the blind via, the volume corresponding to the top diameter, the bottom diameter, and the height; or a taper angle of about 80 degrees or more. The method further includes focusing a Gaussian laser beam emitted from a laser source of the laser drilling system, under laser process parameters, at the substrate surface to remove at least a portion of the mask layer. The method further includes adjusting the laser process parameters based on the one or more properties, and focusing the Gaussian laser beam, under the adjusted laser process parameters, to remove at least a portion of the dielectric layer within the volume to form the blind via.

[0009] In another embodiment, a method of forming a blind via in a substrate is provided. The method includes conveying the substrate to a scanning chamber of a laser drilling system, the substrate comprising a conductive layer of about 2 μm or more in height, a dielectric layer disposed on at least a portion of the conductive layer, and a pre-etched mask layer disposed on at least a portion of the dielectric layer, the pre-etched mask layer having blind via openings to expose at least a portion of the dielectric layer, the at least a portion of the dielectric layer providing a substrate surface. The method further includes detecting the blind via openings of the pre-etched mask layer by a sensor of the laser drilling system, and determining one or more properties of the blind via, the one or more properties of the blind via comprising: a top diameter of the blind via and a bottom diameter of the blind via, the blind via having a height from the top diameter to bottom diameter, the top diameter being greater than the bottom diameter, and the top diameter corresponding to the blind via openings; a volume of the blind via, the volume corresponding to the top diameter, the bottom diameter, and the height; or a taper angle of 80 degrees or more. The method further includes focusing a Gaussian laser beam emitted from a laser source of the laser drilling system, under laser process parameters, at the substrate surface to remove a first portion of the dielectric layer within the volume of the blind via without causing more than half of the thickness of mask layer damage to the pre-etched mask layer. The method further includes adjusting the laser process parameters based on the one or more properties, and focusing the Gaussian laser beam, under the adjusted laser process parameters, to remove a second portion of the dielectric layer within the volume to form the blind via.

[0010] In an embodiment, an apparatus for forming a blind via in a substrate is provided. The apparatus includes an optical device comprising a galvanometer scanner having a plurality of reflecting facets and an axis of rotation, and a beam expander and collimator. The apparatus further includes a femtosecond laser beam source configured to direct electromagnetic radiation to the beam expander and configured to emit a Gaussian laser beam, a transport assembly configured to position the substrate to receive the electromagnetic radiation reflected from at least one of the reflecting facets of the galvanometer scanner, and a height sensor configured to detect a height of a mask layer disposed on at least a portion of the substrate. The apparatus further includes a controller configured to receive signals from the height sensor and to control the femtosecond beam laser source and the transport assembly based on the signals received from the height sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

[0012] FIG. 1A is an example substrate according to at least one embodiment of the present disclosure.

[0013] FIG. 1B is an example substrate according to at least one embodiment of the present disclosure.

[0014] FIG. 1C is an example substrate with a blank mask layer according to at least one embodiment of the present disclosure.

[0015] FIG. 1D illustrates an example substrate with a pre-etched mask layer according to at least one embodiment of the present disclosure.

[0016] FIG. 2A is an example substrate after forming a blind via according to at least one embodiment of the present disclosure.

[0017] FIG. 2B is an example substrate after forming a blind via in a substrate having a blank mask layer according to at least one embodiment of the present disclosure.

[0018] FIG. 2C is an example substrate after forming a blind via in a substrate having a pre-etched mask layer according to at least one embodiment of the present disclosure.

[0019] FIG. 3 is an example method of forming a blind via in a substrate according to at least one embodiment of the present disclosure.

[0020] FIG. 4 is an example method of forming a blind via in a substrate according to at least one embodiment of the present disclosure.

[0021] FIG. 5 is a schematic plan view of an example substrate having a plurality of blind vias formed by apparatus and methods disclosed herein according to at least one embodiment of the present disclosure.

[0022] FIG. 6A is a schematic side view of an example laser drilling system according to at least one embodiment of the present disclosure.

[0023] FIG. 6B is an enlarged side view of the example optical device of FIG. 6A according to at least one embodiment of the present disclosure.

[0024] FIG. 7 is an isometric view of an example laser drilling tool having the example laser drilling system of FIG. 6A disposed therein according to at least one embodiment of the present disclosure.

[0025] FIG. 8 is a side view of an example embodiment of the optical alignment device of FIG. 7 according to at least one embodiment of the present disclosure.

[0026] FIG. 9 is an isometric view of the example optical alignment device of FIG. 8 according to at least one embodiment of the present disclosure.

[0027] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the Figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0028] Embodiments of the present disclosure generally relate to methods and apparatus for drilling blind vias. The inventors have discovered a method and apparatus to laser drill blind vias in a panel that includes a conductive layer and an insulating layer. Unlike conventional laser drilling methods and apparatus, the method and apparatus described herein can enable formation of blind vias having taper angles greater than about 80° , and can enable formation of blind vias having diameters of less than about $40\text{ }\mu\text{m}$, such as about $5\text{ }\mu\text{m}$ to about $10\text{ }\mu\text{m}$. The methods and apparatus herein enable formation of quality blind vias for high-volume manufacturing.

[0029] Alongside the trend toward smaller diameter blind vias, e.g., diameters from $5\text{ }\mu\text{m}$ to $10\text{ }\mu\text{m}$, for higher interconnect density, certain blind via quality parameters remain. These include via top diameter and roundness, via bottom diameter and roundness, via taper angle (80° or more), and via pad cleanliness. However, conventional laser-drilling technologies cannot form quality blind vias having such small diameters for high volume manufacturing.

[0030] Conventional approaches for laser drilling blind vias use nanosecond-pulsed 355 nm

ultraviolet (UV) lasers or pulsed CO₂ lasers to directly drill blind vias in panels. These approaches have certain limitations. First, the Gaussian laser beam profile emitted by the laser sources requires complex beam-shaping optics to convert the beam into a top-hat shaped beam profile. Second, the beam-shaping optics employed result in about 40% optical energy loss. Third, the intensity profile of the top-hat shaped beam has a very short depth-of-field (DOF) within which the laser beam profile remains equal in intensity. Here, the intensity profile of the output top-hat beam deforms/degrades quickly from the image plane along the beam propagation axis because of its non-uniform phase distribution, which limits its application in imaging with large DOF. A smaller diameter top-hat image beam has a shorter DOF, and as the DOF becomes shorter than the variation range of panel thickness/chuck flatness, it is difficult to ensure that the top-hat beam always intersects the panel surface on each via drilling location, so the via quality is inconsistent, particularly for high-volume manufacturing. This inability to achieve consistent via quality at higher throughput is not cost-effective. Another conventional method employs a deep ultraviolet (DUV) light excimer laser with a wavelength of 193 to 308 nm. A mask projection converts the laser beam to a top-hat shaped beam profile. This method also has certain limitations. First, the equipment, e.g., laser and optics, and maintenance thereon can be expensive. Second, the intensity profile of the top-hat shaped beam has a very short DOF, thereby limiting its application. Third, the mask is consumable. The methods and apparatus described herein can eliminate (or at least mitigate) the aforementioned deficiencies of conventional approaches to laser drill blind vias.

Substrate

[0031] FIG. 1A is an example substrate **100** according to at least one embodiment. The example substrate **100** can be utilized for structural support and electrical interconnection of semiconductor packages. The example substrate **100** generally includes a core structure **102**, a conductive layer **104**, and an insulating layer **106**.

[0032] In at least one embodiment, the core structure **102** includes a patterned (e.g., structured) substrate formed of any suitable substrate material. For example, the core structure **102** can include a substrate formed from a III-V compound semiconductor material, silicon, crystalline silicon (e.g., Si<100> or Si<111>), silicon oxide, silicon germanium, doped or undoped silicon, doped or undoped polysilicon, silicon nitride, quartz, glass (e.g., borosilicate glass), sapphire, alumina, and/or ceramic materials. In at least one embodiment, the core structure **102** includes a monocrystalline p-type or n-type silicon substrate. In some embodiments, the core structure **102** includes a polycrystalline p-type or n-type silicon substrate. In another embodiment, the core structure **102** includes a p-type or an n-type silicon solar substrate. The substrate utilized to form the core structure **102** can further have a polygonal or circular shape. For example, the core structure **102** can include a substantially square silicon substrate having lateral dimensions from about 120 to and about 180 mm, with or without chamfered edges. In another example, the core structure **102** includes a circular silicon-containing wafer having a diameter from about 20 mm to about 700 mm, such as from about 100 mm to about 50 mm, for example about 300 mm.

[0033] The conductive layer **104** is formed on one or more surfaces of the core structure **102**. The conductive layer **104** can be formed of a metallic material, such as copper (Cu), tungsten (W), chromium (Cr), molybdenum (Mo), aluminum (Al), gold (Au), nickel (Ni), palladium (Pd), or the like, or a combination thereof. In at least one embodiment, the conductive layer includes a layer of tungsten on copper. In at least one embodiment, the conductive layer **104** has a height $H_{sub.0}$ that is about 100 μm or less, such as about 50 μm or less, such as 25 μm or less. For example, the conductive layer **104** can have a height $H_{sub.0}$ from about 5 μm to about 20 μm , such as a height $H_{sub.0}$ from about 7 μm to about 18 μm , such as from about 10 μm to about 15 μm . In at least one embodiment, the height $H_{sub.0}$ ranges from $H_{sub.0a}$ to $H_{sub.0b}$, where $H_{sub.0a}$ to $H_{sub.0b}$ can be, independently, e.g., about 1 μm , about 2 μm , about 3 μm , about 4 μm , about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 11 μm , about 12 μm , about 13 μm , about 14 μm , about 15 μm , about 16 μm , about 17 μm , about 18 μm , about 19 μm , or about 20 μm , as

long as $H_{\text{sub.0a}} < H_{\text{sub.0b}}$.

[0034] The insulating layer **106** is formed on one or more surfaces of the conductive layer **104**. In at least one embodiment, the insulating layer **106** is formed of polymer-based dielectric materials. For example, the insulating layer **106** is formed from a flowable build-up material, typically in the form of a dry film. Accordingly, although hereinafter referred to as an “insulating layer,” the insulating layer **106** can also be described as a dielectric layer. In some embodiments, the insulating layer **106** is formed of an epoxy resin material having a ceramic filler, such as silica (SiO_2) particles. Other examples of ceramic fillers that can be utilized to form the insulating layer **106** include aluminum nitride (AlN), aluminum oxide (Al_2O_3), silicon carbide (SiC), silicon nitride (Si_3N_4), $\text{Sr}_2\text{Ce}_2\text{Ti}_5\text{O}_{16}$, zirconium silicate (ZrSiO_4), wollastonite (CaSiO_3), beryllium oxide (BeO), cerium dioxide (CeO_2), boron nitride (BN), calcium copper titanium oxide ($\text{CaCu}_3\text{Ti}_4\text{O}_{12}$), magnesium oxide (MgO), titanium dioxide (TiO_2), zinc oxide (ZnO) and the like. In some examples, the ceramic fillers utilized to form the insulating layer **106** have particles ranging in size between about 40 nm and about 1.5 μm , such as between about 80 nm and about 1 μm . For example, the ceramic fillers have particles ranging in size between about 200 nm and about 800 nm, such as between about 300 nm and about 600 nm. In some embodiments, the insulating layer is a polymer with or without particle reinforcement. The insulating layer can be a dry dielectric film or a liquid dielectric film.

[0035] In at least one embodiment, the insulating layer **106** has a height $H_{\text{sub.1}}$ that is about 100 μm or less, such as about 50 μm or less, such as 25 μm or less. For example, the insulating layer **106** can have a height $H_{\text{sub.0}}$ from about 5 μm to about 20 μm , such as a height $H_{\text{sub.1}}$ from about 7 μm to about 18 μm , such as from about 10 μm to about 15 μm . In at least one embodiment, the height $H_{\text{sub.1}}$ ranges from $H_{\text{sub.1a}}$ to $H_{\text{sub.1b}}$, where $H_{\text{sub.1a}}$ to $H_{\text{sub.1b}}$ can be, independently, e.g., about 1 μm , about 2 μm , about 3 μm , about 4 μm , about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 11 μm , about 12 μm , about 13 μm , about 14 μm , about 15 μm , about 16 μm , about 17 μm , about 18 μm , about 19 μm , or about 20 μm , as long as $H_{\text{sub.1a}} < H_{\text{sub.1b}}$.

[0036] FIG. 1B is an example substrate **150** according to at least one embodiment. The example substrate **150** can be utilized for structural support and electrical interconnection of semiconductor packages. The example substrate **150** generally includes a core structure **152**, conductive layers **154a**, **154b** formed on opposing surfaces of the core structure **152**, an insulating layer **156a** formed on a surface of the conductive layer **154a**, and an insulating layer **156b** formed on a surface of the conductive layer **154b**. Example properties and characteristics of the core structure **152**, conductive layers **154a**, **154b**, and insulating layers **156a**, **156b** are similar to those described above with respect to the core structure **102**, conductive layer **104**, and insulating layer **106**, respectively.

[0037] FIG. 1C is an example substrate **170** with a blank mask layer (e.g., not pre-etched) according to at least one embodiment. The example substrate **170** can be utilized for structural support and electrical interconnection of semiconductor packages. The example substrate **170** generally includes a core structure **172**, a conductive layer **174** formed on a surface of the core structure **172**, an insulating layer **176** formed on a surface of the conductive layer **174**, and a mask layer **178** formed on a surface of the insulating layer **176**. Example properties and characteristics of the core structure **172**, conductive layer **174**, and insulating layer **176** are similar to those described above with respect to the core structure **102**, conductive layer **104**, and insulating layer **106**, respectively.

[0038] In at least one embodiment, the mask layer **178** comprises Al, Cu, W, Mo, Cr, or a combination thereof. In some embodiments the mask layer (with or without openings) has a height $H_{\text{sub.2}}$ of about 2 μm or less, such as from about 0.03 μm to about 2 μm or from about 0.05 μm to about 1 μm . In at least one embodiment, the height $H_{\text{sub.2}}$ ranges from $H_{\text{sub.2a}}$ to $H_{\text{sub.2b}}$, where $H_{\text{sub.2a}}$ to $H_{\text{sub.2b}}$ can be, independently, e.g., about 0.1 μm , about 0.2 μm , about 0.3 μm , about 0.4 μm , about 0.5 μm , about 0.6 μm , about 0.7 μm , about 0.8 μm , about 0.9 μm , about 1 μm ,

about 1.1 μm , about 1.2 μm , about 1.3 μm , about 1.4 μm , about 1.5 μm , about 1.6 μm , about 1.7 μm , about 1.8 μm , about 1.9 μm , or about 2 μm , as long as $H.\text{sub}.2a < H.\text{sub}.2b$.

[0039] The mask layer **178** can be formed by, e.g., deposition, sputter, or electrical plating. In at least one embodiment, the mask layer is laser ablated. In some embodiments, the mask layer **178** is removed by an etching operation after the blind via has been formed by processes described herein. In some embodiments, such as when the mask layer **178** is Mo and/or W, the mask is maintained after the blind via has been formed and is combined with a to-be-deposited conductive layer (e.g., Cu) to function as an interconnect layer.

[0040] In at least one embodiment, and referring again to FIG. **1B**, the example substrate includes a mask on a surface of insulating layer **156a** and/or a surface of insulating layer **156b**. The mask layer can be blank or pre-etched to form via openings. Characteristics of each mask layer are similar to that described above in relation to mask layer **178**.

[0041] FIG. **1D** is an example substrate **180** with a pre-etched (or pre-opened) mask layer according to at least one embodiment. Here, the example substrate **180** is pre-etched to form via openings. The example substrate **180** can be utilized for structural support and electrical interconnection of semiconductor packages. The example substrate **180** generally includes a core structure **182**, a conductive layer **184** formed on a surface of the core structure **182**, an insulating layer **186** formed on a surface of the conductive layer **184**, and a mask layer **188** formed on a surface of the insulating layer **186**. Example properties and characteristics of the core structure **182**, conductive layer **184**, and insulating layer **186** are similar to those described above with respect to the core structure **102**, conductive layer **104**, and insulating layer **106**, respectively. Example properties and characteristics of the mask layer **188** are similar to those of mask layer **178** described above.

[0042] FIG. **2A** is an example substrate **200** after forming a blind via in the example substrate **100** (shown in FIG. **1A**) according to at least one embodiment. The example substrate **200** generally includes a core structure **202**, a conductive layer **204**, and an insulating layer **206**. The blind via **208** extends from a surface of the insulating layer **206** to the conductive layer **204**. The blind via **208** has a top diameter $D.\text{sub}.1$, a bottom diameter $D.\text{sub}.2$, a taper angle ($A.\text{sub}.1$), and a volume $V.\text{sub}.1$. In some embodiments, the blind via **208** penetrates, or minimally penetrates, the conductive layer **204**. In some embodiments, the blind via does not penetrate, or substantially does not penetrate, the conductive layer **204**.

[0043] The diameters of the top diameter $D.\text{sub}.1$ and the bottom diameter $D.\text{sub}.2$ can be about 20 μm or less, such as about 15 μm or less, such as from about 5 μm to about 10 μm . In at least one embodiment, the top diameter $D.\text{sub}.1$ ranges from $D.\text{sub}.1a$ to $D.\text{sub}.1b$, where $D.\text{sub}.1a$ to $D.\text{sub}.1b$ can be, independently, e.g., about 1 μm , about 2 μm , about 3 μm , about 4 μm , about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 11 μm , about 12 μm , about 13 μm , about 14 μm , about 15 μm , about 16 μm , about 17 μm , about 18 μm , about 19 μm , or about 20 μm , as long as $D.\text{sub}.1a < D.\text{sub}.1b$. In at least one embodiment, the bottom diameter $D.\text{sub}.2$ ranges from $D.\text{sub}.2a$ to $D.\text{sub}.2b$, where $D.\text{sub}.2a$ to $D.\text{sub}.2b$ can be, independently, e.g., about 1 μm , about 2 μm , about 3 μm , about 4 μm , about 5 μm , about 6 μm , about 7 μm , about 8 μm , about 9 μm , about 10 μm , about 11 μm , about 12 μm , about 13 μm , about 14 μm , about 15 μm , about 16 μm , about 17 μm , about 18 μm , about 19 μm , or about 20 μm , as long as $D.\text{sub}.2a < D.\text{sub}.2b$.

[0044] In some embodiments, the bottom diameter $D.\text{sub}.2$ is less than the top diameter $D.\text{sub}.1$, such that the blind via has a taper. The taper corresponds to a ratio of the bottom diameter $D.\text{sub}.2$ to top diameter $D.\text{sub}.1$. In at least one embodiment, the ratio of the bottom diameter $D.\text{sub}.2$ to the top diameter $D.\text{sub}.1$ is from about $0.353 \cdot H.\text{sub}.IL$ (where $H.\text{sub}.IL$ is the height of the insulating layer, e.g., insulating layer **206**) to about 1, such as from about $0.4 \cdot H.\text{sub}.1$ to about 1, such as from about $0.5 \cdot H.\text{sub}.1$ to about 1.

[0045] In at least one embodiment, the blind via **208** has a taper angle $A.\text{sub}.1$ which corresponds to the angle between the interior wall of the blind via **208** and a surface of the conductive layer **204**.

In at least one embodiment, the taper angle A.sub.1 is from about 75° to 90°, such as from about 80° to about 89°. In at least one embodiment, the taper angle A.sub.1 ranges from A.sub.1a to A.sub.1b, where A.sub.1a to A.sub.1b can be, independently, e.g., about 71°, about 72°, about 73°, about 74°, about 75°, about 76°, about 77°, about 78°, about 79°, about 80°, about 81°, about 82°, about 83°, about 84°, about 85°, about 86°, about 87°, about 88°, about 89°, or about 90°, as long as $A_{sub.1a} < A_{sub.1b}$.

[0046] FIG. 2B is an example substrate **250** after forming a blind via in the example substrate **170** having a blank mask (shown in FIG. 1C) according to at least one embodiment. The example substrate **250** generally includes a core structure **252**, a conductive layer **254**, an insulating layer **256**, and a mask layer **260**. The blind via is shown by reference numeral **258**. Characteristics of core structure **252**, conductive layer **254**, insulating layer **256**, height H.sub.0, height H.sub.1, diameter D.sub.1, diameter D.sub.2, and taper angle A.sub.1 are similar to those described above. Angle A.sub.2 can correspond substantially to taper angle A.sub.1 and/or can have similar characteristics of taper angle A.sub.1. In at least one embodiment, A.sub.2 is equal to or smaller than A.sub.1. In at least one embodiment, the taper angle A.sub.2 ranges from A.sub.2a to A.sub.2b, where A.sub.2a to A.sub.2b can be, independently, e.g., about 71°, about 72°, about 73°, about 74°, about 75°, about 76°, about 77°, about 78°, about 79°, about 80°, about 81°, about 82°, about 83°, about 84°, about 85°, about 86°, about 87°, about 88°, about 89°, or about 90°, as long as $A_{sub.2a} < A_{sub.2b}$.

[0047] In some embodiments, diameter D.sub.3 of mask layer **260** is about equal to diameter D.sub.1 or larger, such as from about diameter D.sub.1 to about diameter D.sub.1+5 μm, such as about diameter D.sub.1+4 μm, about diameter D.sub.1+3 μm, about diameter D.sub.1+2 μm, or about diameter D.sub.1+1 μm. In at least one embodiment, diameter D.sub.3 ≥ 2 diameter D.sub.1 ≥ diameter D.sub.2. The blind via can correspond to reference numeral **258** when the mask layer is removed from the substrate after processing, or can correspond to numeral **262** when the mask layer is maintained on the substrate after processing.

[0048] FIG. 2C is an example substrate **270** after forming a blind via in the example substrate **180** having a pre-etched/pre-opened mask (shown in FIG. 1D) according to at least one embodiment. The example substrate **270** generally includes a core structure **272**, a conductive layer **274**, an insulating layer **276**, and a mask layer **280**. The blind via is shown by reference numeral **278**. Characteristics of core structure **272**, conductive layer **274**, insulating layer **276**, height H.sub.0, height H.sub.1, diameter D.sub.1, diameter D.sub.2, taper angle A.sub.1, and taper angle A.sub.2 are similar to those described above. In at least one embodiment, taper angle A.sub.2 is substantially 90° or corresponds substantially to taper angle A.sub.1. In at least one embodiment, taper angles A.sub.1 and A.sub.2 are, independently, e.g., about 71°, about 72°, about 73°, about 74°, about 75°, about 76°, about 77°, about 78°, about 79°, about 80°, about 81°, about 82°, about 83°, about 84°, about 85°, about 86°, about 87°, about 88°, about 89°, or about 90°. Taper angle A.sub.1 can be equal to, less than, or greater than taper angle A.sub.2. Diameter D.sub.3 (in units of μm) of mask layer **280** can be about (D.sub.1+H.sub.2) or less, such as from about D.sub.1 to about (D.sub.1+(1/4) H.sub.2), such as from about D.sub.1 to about (D.sub.1+(1/3) H.sub.2), such as from about D.sub.1 to about (D.sub.1+(1/2)H.sub.2). In at least one embodiment, diameter D.sub.3 ≥ diameter D.sub.1. The blind via can correspond to reference numeral **278** when the mask layer is removed from the substrate after processing, or can correspond to numeral **282** when the mask layer is maintained on the substrate after processing.

[0049] In some embodiments, the allowable variation range of taper angle A.sub.2 is wider than that of taper angle A.sub.1, e.g., $70^\circ \leq A_{sub.2} \leq 90^\circ$, $75^\circ \leq A_{sub.1} \leq 90^\circ$. In some examples, $80^\circ \leq A_{sub.1}$ for 5-10 μm diameter vias; for larger than 10 μm vias, $75^\circ \leq A_{sub.1}$.

[0050] In at least one embodiment, the mask layer (with or without openings) has a height of about 2 μm or less, such as from about 0.03 μm to about 2 μm or from about 0.05 μm to about 1 μm.

[0051] The mask layers described herein can be opened by photolithography. In some

embodiments, a photoresist layer is applied on top of the metal mask. The photoresist layer can be patterned by photolithography to define the sizes and locations of via openings on the metal mask layer. The opening on the metal mask layer can be formed by etching processes, such as a plasma etch and a wet chemical etch. Lithography-defined openings can allow for relaxed positional accuracies and can permit improved laser throughput.

Processes

[0052] Embodiments described herein also include processes for laser drilling blind vias. The laser source for laser drilling blind vias can be a femtosecond laser. Suitable femtosecond-based laser processes can be characterized by a high peak intensity (irradiance) that usually leads to nonlinear interactions in various materials.

[0053] FIG. 3 shows an example method **300** of forming a blind via in a substrate according to at least one embodiment. In at least one embodiment, example method **300** is used on a substrate having a mask layer with no openings (i.e., blank). The example method **300** includes conveying the substrate to a scanning chamber at operation **302**. Here, the substrate can include a conductive layer (e.g., a copper layer), a dielectric layer disposed on at least a portion of the conductive layer, and a mask layer disposed on at least a portion of the dielectric layer, the mask layer providing a substrate surface. The example method **300** further includes determining one or more properties of the blind via at operation **304**. The one or more properties can include a top diameter of the blind via, a bottom diameter of the blind via, a height of the blind via, a volume of the blind via, and a taper angle of the blind via. In at least one embodiment, the taper angle is greater than 75° or greater than 80° as described above. In some embodiments, the top diameter is greater than the bottom diameter, and the top diameter corresponds to the blind via opening. The volume of the blind via corresponds to the top diameter, the bottom diameter, and the height. The example method **300** further includes focusing a laser beam emitted from a laser source, under laser process parameters, at the substrate surface at operation **306**. This operation can result in removing a first portion of the dielectric layer within the volume, while only causing gentle melting and polishing on the surface of the conductive layer underneath the dielectric layer, with the ablation depth in the conductive layer no more than, e.g., about 2 μm. In some embodiments, operation **306** results in removing about half of the thickness of the mask layer or less. Accordingly, and in some embodiments, the conductive layer has no substantial damage after operation **306**.

[0054] The example method **300** further includes adjusting the laser process parameters based on the one or more properties at operation **308**. In some embodiments, the laser process parameters include a laser power, a laser energy in a burst, a focal beam diameter, a focus height, a burst energy, a pulse energy, a number of pulses in a burst, a pulse frequency, a burst frequency, a beam spot size, an M.sup.2 value, an offset of beam focusing from substrate surface (either above and/or below the substrate surface), or a combination thereof. M.sup.2 is a unitless laser beam spatial characteristic. It measures the difference between an actual laser beam and a Gaussian beam. These and other laser process parameters are described below. The example method **300** further includes focusing the laser beam, under adjusted laser process parameters, to remove at least a portion of the dielectric layer within the volume to form the blind via at operation **310**.

[0055] In some embodiments, further operations are performed such as removing the mask layer from the substrate. In at least one embodiment, and when the mask layer comprises Mo and/or W, a further operation includes depositing a layer of copper on the mask layer.

[0056] FIG. 4 shows an example method **400** of forming a blind via in a substrate according to at least one embodiment. In at least one embodiment, example method **400** is used to form a blind via on a substrate having a pre-etched mask layer. The example method **400** includes conveying the substrate to a scanning chamber at operation **402**. Here, the substrate can include a conductive layer (e.g., a copper layer), a dielectric layer disposed on at least a portion of the conductive layer, and a pre-etched mask layer disposed on at least a portion of the dielectric layer, the pre-etched mask layer having blind via openings to expose at least a portion of the dielectric layer and the dielectric

layer providing a substrate surface. The example method **400** further includes determining one or more properties of the blind via at operation **404**. The one or more properties can include a top diameter of the blind via, a bottom diameter of the blind via, a height of the blind via, a volume of the blind via, and a taper angle of the blind via. In at least one embodiment, the taper angle is greater than 75° or greater than 80° as described above. In some embodiments, the top diameter is greater than the bottom diameter, and the top diameter corresponds to the blind via opening. The volume of the blind via corresponds to the top diameter, the bottom diameter, and the height. The example method **400** further includes focusing a laser beam emitted from a laser source, under laser process parameters, at the substrate surface at operation **406**. This operation can result in removing a first portion of the dielectric layer within the volume, and can be performed without causing substantial damage to the pre-etched mask layer and/or without causing substantial damage to the conductive layer. “Without substantial damage to the pre-etched mask layer” refers to slightly melting the mask layer as to “shine/polish” mask surface but no visible ablation induced material removal from mask surface as to cause top via diameter change or induce oval shape of via. The mask is a sacrificial layer that can be removed. In at least one embodiment, the tolerable mask damage depends on the mask height. As a non-limiting example, if the mask height is about $0.3\text{ }\mu\text{m}$, the damage/ablation depth into the mask should be less than about $0.3\text{ }\mu\text{m}$. As another non-limiting example, if the mask height is about $2\text{ }\mu\text{m}$, the damage/ablation depth into the mask should be less than about $2\text{ }\mu\text{m}$.

[0057] The example method **400** further includes adjusting the laser process parameters based on the one or more properties at operation **408**. In some embodiments, the laser process parameters include a laser power, a laser energy in a burst, a focal beam diameter, a focus height, a burst energy, a pulse energy, a number of pulses in a burst, a pulse frequency, a burst frequency, a beam spot size, an M.sup.2 value, an offset of beam focusing from substrate surface, or a combination thereof. These and other laser process parameters are described below. The example method **400** further includes focusing the laser beam, under adjusted laser process parameters, to remove at least a portion of the dielectric layer within the volume to form the blind via at operation **410**.

[0058] In some embodiments, further operations are performed, such as removing the pre-etched mask layer from the substrate. In at least one embodiment, and when the pre-etched mask layer comprises Mo and/or W, a further operation includes depositing a layer of copper on the pre-etched mask layer.

[0059] The femtosecond laser source (such as an ultraviolet laser) used in at least some embodiments herein has a number of adjustable characteristics (laser process parameters) as described below. In at least one embodiment, the laser process parameters include one or more of the following characteristics: [0060] (1) The femtosecond laser source has a pulse width or pulse width range from about 1 femtosecond (fs) to about 1000 fs, such as from about 100 fs to about 750 fs, such as from about 200 fs to about 500 fs. [0061] (2) The femtosecond laser source has a wavelength or wavelength range from about 250 nanometers (nm) to about 2000 nm, such as from about 266 nm to about 1500 nm, such as from about 350 nm to about 540 nm. In at least one embodiment, the femtosecond laser source has a wavelength of about 400 nm or lower. [0062] (3) The femtosecond laser source and corresponding optical system provide a focal spot or focal spot range at the work surface in the range from about 1.5 microns (μm) to about 12 μm , such as from about 3 μm to about 10 μm , such as from about 4 μm to about 8 μm . The spatial beam profile at the work surface can be a single mode (Gaussian) profile. [0063] (4) The femtosecond laser source outputs bursts of pulses. Within each burst, the pulse frequency or range of pulse frequency is about 500 MHz or more, such as about 1 GHz or more, such as 2 GHz or more, such as from about 1 GHz to about 10 GHz, such as from about 2 GHz to about 9 GHz, such as from about 3 GHz to about 8 GHz, such as from about 4 GHz to about 7 GHz, such as from about 5 GHz to about 6 GHz. In at least one embodiment, the pulse frequency or range of pulse frequency within each burst is from about 2 GHz to about 5 GHz. [0064] (5) The number of pulses in each burst outputted from

the femtosecond laser source can be adjusted. The number of pulses in each burst can be about 2 or more, or about 3 or more, such as from about 5 to about 100, such as from about 10 to about 100, such as from about 20 to about 90, such as from about 40 to about 80, such as from about 50 to about 70, such as from about 55 to about 65. In at least one embodiment, the number of pulses in each burst is from about 20 to about 100. [0065] (6) The burst frequency can be adjusted. The burst frequency of the femtosecond laser source can be about 100 kHz or more, such as about 500 KHz or more. In at least one embodiment, the burst frequency is from about 200 kHz to about 5 MHz or from about 500 KHz to about 5 MHz, such as from about 300 KHz to about 2 MHz, from about 1 MHz to about 2 MHz, or from about 500 kHz to about 1 MHz. [0066] (7) The femtosecond laser source delivers an amount of laser energy in a burst from about 1 μ J to about 100 pJ, such as from about 1 μ J to about 80 pJ, such as from about 3 μ J to about 50 pJ, such as from about 5 μ J to about 20 μ J. In at least one embodiment, the laser and the burst frequency are set to achieve an amount of laser energy in a burst that can be about 5 J or more, such as from about 5 μ J to about 50 μ J, such as from about 10 μ J to about 30 pJ. [0067] (8) The laser power of the femtosecond laser source is about 1 W or more, such as from about 1 W to about 100 W, such as from about 5 W to about 80 W, such as from about 10 W to about 50 W. [0068] (9) The M.sup.2 value that characterizes laser beam quality is about 1.5 or less (M.sup.2 is always ≥ 1.0), such as from about 1 to about 1.3, such as from about 1.1 to about 1.2, or from about 1 to about 1.15, or from about 1 to about 1.1. [0069] (10) The focal beam diameter can be from about 2 μ m to about 10 μ m, such as from about 3 μ m to about 6 μ m for drilling 5 μ m diameter vias. The focal beam diameter can be from about 7 μ m to about 12 μ m, such as from about 8 to about 11 μ m for drilling 10 μ m diameter vias. The focal beam diameter is the laser beam spot diameter at the work surface, which is the result of the output laser beam going through a beam expander and then being focused by a focusing lens. [0070] (11) The offset of beam focusing from a substrate, which is also called the focus height, is from about 0 μ m to 100 μ m, such as from about 0 μ m to about 50 μ m, such as from about 0 μ m to about 30 μ m. Focus height is an adjustable parameter. In some embodiments, for example, the focal spot is set right on the panel surface to cut through the mask and remove some of the insulating layer material, and then the focus height is adjusted so that the focal plane is above the panel surface—that is, to set the laser beam intersecting panel surface off—focus to get a lower fluence that can only remove the insulating layer material but not damage mask layer and conductive layer at the via bottom. [0071] (12) The number of bursts are about 2 or more, such as from about 5 to about 20. [0072] (13) The focused beam spot size is about 80% to about 120% of the targeted entry diameter of the hole at the dielectric surface (e.g., entry diameter D.sub.1 of the hole to be drilled as defined in FIG. 2C or the diameter D.sub.1 of the hole to be drilled as defined in FIG. 2B), such as from about 90% to about 110%, such as from about 95% to about 100%.

[0073] The focused beam diameter can be adjustable using a programmable beam expander to drill different diameter vias.

[0074] The femtosecond laser source can be an electromagnetic radiation source such as a diode-pumped solid state (DPSS) laser or other similar radiation emitting sources that can provide and emit a continuous or pulsed beam. According to the lasing medium (crystal) configuration, and in at least one embodiment, the DPSS lasers can be a rod crystal laser, a fiber laser, a disc laser, a rod-type photonic crystal fiber laser, an innoslab laser, or a hybrid of them. In some embodiments, the laser source includes multiple laser diodes, each of which produces uniform and spatially coherent light at the same wavelength.

Apparatus

[0075] Embodiments of the present disclosure also generally relate to apparatus for laser drilling blind vias. FIG. 6A is a schematic side view of an example laser drilling system **600** for carrying out certain aspects of the present disclosure according to some embodiments.

[0076] The laser drilling system **600** includes an enclosure **602** having a substrate positioning system **605** therein. The substrate positioning system **605** can be a conveyor for supporting and

transporting substrates **510** through the laser drilling system **600**. The laser drilling system **600** can be used to drill blind vias **501** (FIG. 5) in one or more layers disposed on a substrate **510** (or a plurality of substrates) in accordance with embodiments of the present disclosure. Each substrate **510** can be an example substrate illustrated in FIGS. 1 and 2. In at least one embodiment, the mask of the substrate **510** can face upward for processing in the laser drilling system **600**.

[0077] In at least one embodiment, the substrate positioning system **605** is a linear conveyor system that includes a transport belt **615**, which can be continuous, of a material configured to support and transport a line of the substrates **510** through the laser drilling system **600** in a flow path "A". The enclosure **602** is positioned between a loading station **617A** for providing substrates **510** and an unloading station **617B** for receiving processed substrates **510**. The loading station **617A** and the unloading station **617B** can be coupled to the enclosure **602** and can include robotic equipment and/or transfer mechanisms that provide substrates **510** to the transport belt **615**. The substrate positioning system **605** includes support rollers **620** that support and/or drive the transport belt **615**. The support rollers **620** are driven by a mechanical drive **625**, such as a motor/chain drive, and are configured to transport the transport belt **615** at a linear speed of from about 100 millimeters per second (mm/s) to about 2000 mm/s, such as from about 500 mm/s to about 2000 mm/s, or from about 500 mm/s to about 1500 mm/s during operation. The mechanical drive **625** can be an electric motor, such as an alternating current (AC) or direct current (DC) servo motor. The transport belt **615** can be made of, e.g., stainless steel, a polymeric material, and/or aluminum. In at least one embodiment, the transport belt **615** includes two parallel belts that can be spaced apart in the X-direction, wherein each of the two parallel belts have a width in the X-direction that is smaller than the X-direction dimension of the substrates **510**. In this configuration, each substrate **510** in the laser drilling system **600** can be disposed on a portion of both parallel belts.

[0078] The substrate positioning system **605** can be a transfer device configured to sequentially transport a line of the substrates **510** (e.g., in the flow path "A") toward and through a laser scanning apparatus **630**. The laser scanning apparatus **630** includes an optical device **635A** coupled to a support member **640** that supports the optical device **635A** above the transport belt **615** and substrates **510**. The laser scanning apparatus **630** also includes a scanning chamber **635B** that is fixed in position relative to the transport belt **615** adjacent the optical device **635A** allowing the substrates **510** to pass therethrough on the transport belt **615**.

[0079] FIG. 6B is a side view of the example optical device **635A** of FIG. 6A, wherein the optical device **635A** is rotated 90 degrees from its standard position shown in FIG. 6A for discussion purposes. FIG. 5 is a schematic plan view of a substrate **510** having a plurality of blind vias **501** formed by the optical device **635A** of FIGS. 6A and 6B. The optical device **635A** includes a housing **641** that provides light or electromagnetic radiation that is directed toward the surface of the substrates **510** as the substrates **510** pass through the scanning chamber **635B** on the transport belt **615**. In at least one embodiment, the optical device **635A**, in combination with the movement of the transport belt **615**, is configured to form a pattern (P), as shown in FIG. 5, which can include a pattern of rows (R) and columns (C) of blind vias **501** formed in a substrate **510**. In some embodiments, the optical device **635A** forms the pattern (P) on the substrate **510** in a time period of about 0.5 milliseconds (ms) or less, such as from about 0.01 ms to about 0.1 ms or from about 0.001 ms to about 0.005 ms using an optics system that can provide a pulsed beam that traverses the substrate **510** at a high speed as the substrate **510** is moved on the transport belt **615**. The optical device **635A** also includes a laser source **642**, such as a femtosecond laser source, that emits light or electromagnetic radiation through an optics system that provides, e.g., about 50000 blind vias in the substrate **510** having desired characteristics described herein, such as diameter and taper angle.

[0080] In laser machining processes, unlike typical plasma process in semiconductor processing, the plasma is more like a cloud or plume that can cover the whole wafer surface, a laser beam, particularly when being focused, is only a tiny spot, therefore, typically a relative movement

between laser beam and sample (e.g., wafer) is needed to process a whole sample. The relative movement can be realized in different ways. In a first way, and in some embodiments, the laser beam is kept stationary and the substrate is moved by a linear stage with motion in X/Y/Z (height)/Theta angle (on X-Y plane to rotate sample)/A tilting angle (on X-Z and Y-Z plane to tilt sample) directions. For micromachining, typically, X/Y/Z/theta movement can be enough. In a second way, and in some embodiments, the laser source and/or laser focusing head is mounted on a X/Y/Z/Theta stage.

[0081] In a third way and in some embodiments, the substrate is kept stationary on a chuck, and the laser beam is scanned across the substrate. Optical scanners are used to direct, position, and/or “scan” a laser beam over a desired area of a substrate. In this case, a light beam is refracted, diffracted, and/or reflected by optical scanners to realize its movement on a substrate surface. Generally, there are three types of optical scanners: acousto-optic scanners that deflect a beam using diffraction; electro-optic scanners that deflect a beam using refraction; and mechanical scanners (resonant, polygonal, and galvanometer scanning types) that deflect a beam using reflection.

[0082] A fourth way is hybrid movement of the laser beam and the sample. For example, the laser beam is scanned across a substrate while moving the substrate on the chuck/stage simultaneously or in sequence. Additionally, or alternatively, the scanner is mounted on a one-axis moving stage to move and scan beam.

[0083] In some embodiments, optical scanning technology such as mechanical scanners, such as polygon and/or galvos, are used to move the laser beam from one position to another position on the substrate surface in order to, e.g., drill via arrays at designated locations on the substrate. These mechanical scanners work by rotating a physical mirror that can be coated to reflect any wavelength or combination of wavelengths off the rotating mirror with very high reflectivity- and thus, high optical throughput. The achievable scan angle in optical degrees is double the actual motor rotation angle. Using that physical mirror, a polygonal scanner can be made to scan a beam over very wide angles, but having the limitation of scanning the same pattern over and over again. A polygon scanner performs one-axis scanning. To generate a 2-dimensional pattern, the other axis-movement can be provided by adding a linear stage or a galvo mirror.

[0084] Mechanical galvanometer-based scanners (or galvos), typically involve a physical mirror operated by a motor. Most often, the mirror is attached to the shaft of the motor, but in some designs the mirror and motor may be a single integral unit. Galvo motors can rotate over a range of angles (typically around $\pm 20^\circ$). Galvo motors can also incorporate a high-precision position detector that provides feedback to a separate controller, delivering pointing repeatability of 5 μ rad (5 mm at a distance of 1 km). Two galvanometer-scanners can be configured for two-axis scanning.

[0085] In at least one embodiment, the speed of the transport belt **615** is controlled at about 100 mm/s to about 5000 mm/s, such as about 250 mm/s to about 2000 mm/s, such as about 0.5 m/s to about 1 m/s, during operation, in order to form multiple blind vias **501** in a substantially linear row in the X-direction (FIG. **6A**) on the substrates **510** as the substrates **510** pass below the optical device **635A** on the transport belt **615** in the Y-direction.

[0086] In at least one embodiment, and as an alternative to (or in addition to transport belt **615**), a galvoscaner is used to change drilling position from one via to the next.

[0087] Referring again to FIG. **6B**, the laser source **642** can emit light or electromagnetic radiation **655** through a process of optical amplification based on stimulated emission of photons. In at least one embodiment, the laser beam emitted by the laser source **642** is a Gaussian beam. In some embodiments, the emitted electromagnetic radiation **655** has a high degree of spatial and/or temporal coherence. In at least one embodiment, the laser source **642** emits a continuous or pulsed wave of light or electromagnetic radiation **655** that is directed to the optics system, which includes a beam expander **644**, a beam collimator **646**, and a galvanometer scanner **650**.

[0088] The galvanometer scanner **650** can include moveable mirrors to steer the laser beam, and

the steering of the beam can be one-, two-, or three-dimensional. In order to position the laser beam in two dimensions, the galvanometer scanner **650** can rotate one mirror along two axes or can reflect the laser beam onto two closely spaced mirrors that are mounted on orthogonal axes. To position the focus of the laser beam in three dimensions, a servo-controlled galvanometer scanner can be used. Characteristics of the laser source **642**, such as a femtosecond laser source, are described above.

[0089] In some embodiments, the pulse width and frequency of the pulses of electromagnetic radiation **655** are controlled by providing the laser source **642** with an external trigger signal that is provided at a desired frequency from a controller **690**.

[0090] The pulses of electromagnetic radiation **655** emitted from the laser source **642** are received at the beam expander **644** having a first diameter, such as from about 1 mm to about 6 mm, such as from about 2 mm to about 5 mm, such as from about 3 mm to about 4 mm. The beam expander **644** can increase the diameter of the electromagnetic radiation **655** to a second diameter by a preset magnification, such as about $2\times$ magnification, such as about $5\times$ magnification, such as about $8\times$ magnification, or the beam expander **644** can have an adjustable range of magnification, such as from about $1\times$ to about $8\times$. The pulses of electromagnetic radiation **655** are then delivered to the beam collimator **646** for narrowing the beam.

[0091] In at least one embodiment, the beam collimation function is integrated into the beam expander **644**. That is, in such embodiments, the beam expander **644** is also itself a beam collimator. Here, the beam expander **644** has at least two functions-expanding to beam size by certain magnification AND collimating the beam. The beam expander can have a fixed magnification (e.g., $2\times$, then the output beam diameter is equal to $2\times$ input beam diameter) or an adjustable range of magnification (e.g., $3\times$ to $8\times$). For example, a 2 mm diameter laser beam can enter the input side of a $3\times$ beam expander, then at the output (exit) side of beam expander is a 6 mm diameter beam as output. This 6 mm beam (as it propagates along the optical axis) can be a divergent beam, or a convergent beam, but typically not, a well collimated beam. A perfectly collimated beam neither diverges nor converges as it propagates along the optical axis. Through the operation of “beam collimating” the beam can be adjusted to have very minor divergence or convergence so that over a few meters travel distance, the beam diameter may only change by about 1% to about 2%, or less.

[0092] From the beam collimator **646**, the pulses of electromagnetic radiation **655** are delivered to the galvanometer scanner **650**, which steers the pulses of electromagnetic radiation **655** through a focusing lens **652** and onto the substrate **510**. The focusing lens **652** can be a telecentric focusing lens. The focusing lens **652** can have one or more lenses.

[0093] The galvanometer scanner **650** steers the pulses of electromagnetic radiation through the focusing lens **652**, which is part of the optics system of the optical device **635A**, and onto a surface of the substrate **510**, which is continually moving in the Y-direction on the transport belt **615** in the scanning chamber **635B** (FIG. 6A). Thus, the transport belt **615** does not need to stop/start during the blind via formation process on the substrate **510**, which can increase throughput. However, in some embodiments, the surface of the substrate is periodically moving in the Y-direction on the transport belt **615** in the scanning chamber **635B** (FIG. 6A). The galvanometer scanner **650** can include a mirror having multiple reflecting facets arranged such that each of the reflecting facets **653** can be generally angled relative to another one of the reflecting facets **653** in a direction relative to an axis of rotation **651** of the galvanometer scanner **650** (into the page in the X direction in FIG. 6B). The angle of each of the reflecting facets **653** of the galvanometer scanner **650** allows the electromagnetic radiation **655** to be scanned in one direction (X-direction in FIG. 6A) across the surface of the substrate **510** as the galvanometer scanner **650** is rotated about the axis of rotation **651** by an actuator **654**. The actuator **654** can be used to control the speed of rotation of the galvanometer scanner **650** to a desired linear speed, such as a speed from about 0.5 m/sec to about 10 m/sec, such as from about 1 m/sec to about 6 m/sec, such as from about 2 m/sec to about 5

m/sec. The speed of scanning can be changed/fixed during the laser drilling process for creation of a pattern on the substrate **510**. In at least one embodiment, the scanning speed is fixed such that all pulses are directed to one spot when drilling an individual via. This is often called percussion drilling or punch. From via to via, the galvo mirror can scan to change the drilling location.

[0094] For example, the rotational speed of the galvanometer scanner **650** can be set at a first speed for creation of a first pattern on one or more first substrates, and the first speed can be maintained during the ablation of each of the one or more first substrates. If a different pattern is desired on one or more second substrates, the rotational speed of the galvanometer scanner **650** can be set at a second speed that is different than the first speed, and the second speed can be maintained during the ablation of each of the one or more second substrates.

[0095] In some embodiments, the rotation of a single facet of the galvanometer scanner **650**, as it is reflecting the delivered pulses of electromagnetic radiation **655** from the laser source **642**, creates a full row (R) of blind vias **501** (e.g., a row in the X-direction) in one or more layers formed on the substrate **510**. The electromagnetic radiation **655** is scanned across the surface of the substrate **510** by use of the galvanometer scanner **650**, while the substrate **510** is transferred in an orthogonally-oriented Y-direction resulting in rows (R) of blind vias **501** (e.g., in the X-direction) spanning the length of the substrate **510** (e.g., in the Y-direction). In another example, the Y-direction is positioned at an angle to the X-direction. In yet another example, the Y-direction is positioned at an angle of about 90 degrees plus or minus a few degrees relative to the X-direction.

[0096] In at least one embodiment, the optics system of the optical device **635A** are configured to deliver a focused beam diameter from about 1.5 μm to about 7 μm , such as from about 2 μm to about 6 μm , such as from about 3 μm to about 5 μm , for formation of the blind vias **501** with an entry diameter D.sub.1 equal to about 5 μm ; and to deliver a focused beam diameter from about 5 μm to about 14 μm , such as from about 7 μm to about 12 μm , such as from about 8 μm to about 10 μm , for formation of the blind vias **501** with an entry diameter D.sub.1 equal to about 10 μm (e.g., D.sub.1 as defined in FIG. 2B or FIG. 2C). In some embodiments, the number of pulse bursts to drill a hole can be from about 1 burst per via to about 40 bursts per via, such as from about 3 bursts per via to about 30 bursts per via, such as from about 5 bursts per via to about 20 bursts per via.

[0097] Referring again to FIG. 6A, the laser drilling system **600** also includes a substrate sensing system **660** including one or more substrate position sensors. The substrate sensing system **660** uses an optical sensor **662** to detect a leading edge **665** of the substrate **510** and sends corresponding signals to a controller **690**. The controller **690**, in turn, sends signals to the optical device **635A** to time the operation of the laser source **642** and the rotation of the galvanometer scanner **650** to begin the laser scanning operation as the leading edge **665** of the substrate **510** is beneath the focusing lens **652**. The controller **690** further controls the rotational speed of the galvanometer scanner **650** to scan a row (R) of blind vias **501** in one or more layers disposed on the substrate **210** as each facet of the galvanometer scanner **650** is rotated across the pulses of electromagnetic radiation **655**. The controller **690** further controls the speed of the substrate positioning system **605** and the rotation of the galvanometer scanner **650**, such that as a first row (R) of blind vias **501** (e.g., aligned in the X-direction) is finished, the next row (R) of blind vias **501** can begin at a desired spacing (e.g., in the direction A) from the first row due to the linear movement of the substrate **510** by the substrate positioning system **605**. Accordingly, and in some embodiments, as the substrate **510** is moved beneath the optical device **635A**, rows (R) of blind vias **501** can be formed in one or more layers of the substrate **510** across the entire width and length of the substrate **510**. The controller **690** further controls the timing of the optical device **635A**, such that as a trailing edge **670** of the substrate **510** passes beneath the focusing lens **652**, the scanning operation ceases after a desired period of time has elapsed until the leading edge of the next substrate **510** is positioned beneath the focusing lens **652**. The controller **690** can be any controller having a suitable processor, software, and memory for operation of the laser drilling system **600**. The substrate sensing system **660** also includes a substrate alignment device **680** configured to

align the substrates **510** prior to entry into the scanning chamber **635B**.

[0098] The controller **690** generally includes a central processing unit (CPU) (not shown), memory (not shown), and support circuits (not shown). The CPU can be one of any form of computer processor used in industrial settings for controlling system hardware and processes. The memory can be connected to the CPU and can be one or more of a readily available memory, such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, or any other form of digital storage, local or remote. Software instructions and data can be coded and stored within the memory for instruction the CPU. The support circuits can also be connected to the CPU for supporting the processor in a conventional manner. The support circuits can include cache, power supplies, clock circuits, input/output circuitry subsystems, and the like. A program (e.g., instructions) readable by the controller **690** can include code to perform tasks relating to monitoring, executing, and controlling the movement, support, and positioning of the substrates **510** along with various process recipe tasks to be performed in the laser drilling system **600**. In at least one embodiment of the controller **690**, the process of forming holes on a surface of a substrate includes at least one scan (to form a full row (R) of blind vias **501** (shown in FIG. 2C)) that can be interrupted at desirable times to further control formation of blind vias **501** in a desirable location on the surface of the substrate. This capability can enable brief and/or selective stopping/starting the train of pulses, and enable advanced patterning features.

[0099] The controller **690** further controls a height sensor **664**. Although FIG. 6A shows the height sensor **664** as being part of the substrate sensing system **660**, the height sensor **664** can be a separate unit. During use of the laser drilling system **600**, the height sensor **664** adjusts the optical device **635A** about the Z-direction before, during, or after ablation as described in FIG. 8 and FIG. 9. In at least one embodiment, the height sensor **664** precisely determines the surface of the mask disposed on the one or more layers of the substrate **510**.

[0100] FIG. 7 is an isometric view of an example laser drilling tool **700** having the optical device **635A** and the scanning chamber **635B** disposed therein. The laser drilling tool **700** includes a main frame **701** having a first side **702A** that can be coupled to a loading station **617A** (shown in FIG. 6A) and a second side **702B** that can be coupled to an unloading station **617B** (shown in FIG. 6A). The main frame **701** can include panels **705** that can function as doors or removable sheets, and a portion of the panels **705** are not shown to illustrate components within the laser drilling tool **700**. The panels **705** include a view window **710** to provide visual access to the interior of the laser drilling tool **700**. The view window **710** can include laser safety glass and/or filters that enable viewing of the electromagnetic radiation during a laser drilling process within the laser drilling tool **700** without the need for safety glasses. Power sources and control equipment, such as a laser power supply **715** (shown in dashed lines) is housed within the main frame **701**. Additionally, an optical alignment device **720** (FIGS. 7 and 8) is coupled to the main frame **701** within the laser drilling tool **700**. The optical alignment device **720** can adjust the position of the optical device **635A** relative to the motion direction of the substrates **510** on the substrate positioning system **605**, thereby adjusting the beam path emitted relative to a substrate.

[0101] As discussed above, the optical device **635A** can be adjusted about the Z-direction before, during, or after ablation using a height sensor **664** as described in FIG. 8 and FIG. 9.

[0102] FIG. 8 is a side view of an example embodiment of the optical alignment device **720**. The optical alignment device can be coupled to a height sensor **664** (not shown). The optical alignment device **720** includes a base plate **800** that is coupled to one or more support members **805** of the main frame **701**. The base plate **800** is movably coupled to a first support plate **810** having a second support plate **815** extending therefrom in a plane substantially orthogonal to the plane of the first support plate **810**. The second support plate **815** generally supports the optical device **635A**. The first support plate **810** is coupled to the base plate **800** by a plurality of adjustment devices **820**, which can include fasteners, linear guides, or a combination thereof. The adjustment devices **820** can enable at least height adjustment (in the Z-direction) of the optical device **635A** and can enable

a theta adjustment in the X-Z plane and/or the Y-Z plane. The height adjustment can be used to adjust the focal length of the focusing lens **652** (FIGS. **6A** and **6B**) of the optical device **635A**. The adjustment devices **820** can also be used to level the second support plate **815** relative to a plane of the transport belt **615**. An adjustable aperture device **840** is provided between the second support plate **815** and the scanning chamber **635B**. The adjustable aperture device **840** can be, e.g., a telescoping housing having an aperture formed therein that is sized to receive a beam path provided by the optical device **635A**. The telescoping housing can be adjusted upward or downward based on a height adjustment of the optical alignment device **720**.

[0103] In some embodiments, the optical alignment device **720** also includes an adjustable mount plate **825** disposed between the second support plate **815** and a lower surface of the optical device **635A**. The adjustable mount plate **825** is secured to the lower surface of the optical device **635A** and is fastened to the second support plate **815** by fasteners **830**. The adjustable mount plate **825** can be adjusted for different angular orientations as well as leveling the optical device **635A** to tune a scan plane **835** of a beam path emitted by the optical device **635A** during processing. As described in more detail in FIG. **9**, the adjustable mount plate **825** can be rotated about the scan-plane axis, e.g., the Z-direction, to adjust the orientation of the scan plane **835** (e.g., plane aligned to the row (R) direction on the substrate in FIG. **5**) of the output of the optical device **635A**. Adjustment of the adjustable mount plate **825** can be performed to alter the beam path(s) within the scan plane of the optical device **635A** to align a row (R) of blind vias **501** on a substrate **510** (FIG. **5**).

[0104] FIG. **9** is an isometric view of the example optical alignment device **720** of FIG. **8**. Although the optical device **635A** is not shown in this view, the scan plane **835** of the optical device **635A** is shown. The adjustable mount plate **825** includes a plurality of slots **900** that receive the fasteners **830** shown in FIG. **8**. Each of the slots **900** can permit the adjustable mount plate **825** to rotate relative to the Z axis in order to adjust the scan plane of the optical device **635A**. For example, a first alignment position **905** of the adjustable mount plate **825** can include a direction wherein the scan plane is substantially parallel to the leading edge **665** of the substrate **510** (shown in FIG. **2A**). In at least one embodiment, the adjustable mount plate **825** is adjusted to a second alignment position **910**, which corresponds to an angle **915**. The angle **915** can be adjusted based on, e.g., the speed of the substrate(s) **510** on the transport belt **615** and/or the scan speed of the electromagnetic radiation **655**. The scan speed of the electromagnetic radiation **655** can be at least partially based on pulse width of the electromagnetic radiation **655** and/or movement of the galvanometer scanner **650** (shown in FIG. **6B**).

[0105] In at least one embodiment, the angle **915** is from about -20° degrees to about $+20^{\circ}$ degrees from the normal axis of the mirror surface, when the speed of the transport belt **615** is about 140 mm/s to about 180 mm/s, with a pulse width of about 1 fs to about 1.5 ms, and the scan speed of the galvanometer scanner **650** is about 1,000 RPM, which results in rows (R) of blind vias **501** that are substantially linear and/or parallel with the leading edge **665** of the substrate **510**.

[0106] Any of the operations described above can be included as instructions in a computer-readable medium for execution by a control unit (e.g., controller **690**) or any other processing system. The computer-readable medium can comprise any suitable memory for storing instructions, such as read-only memory (ROM), random access memory (RAM), flash memory, an electrically erasable programmable ROM (EEPROM), a compact disc ROM (CD-ROM), a floppy disk, and the like.

[0107] For the sake of brevity, only certain ranges are explicitly disclosed herein. However, ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited, as well as, ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited, in the same way, ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, within a range includes every point or individual value between its end points even though not explicitly recited. Thus,

every point or individual value may serve as its own lower or upper limit combined with any other point or individual value or any other lower or upper limit, to recite a range not explicitly recited. [0108] For the purposes of this present disclosure, and unless otherwise specified, all numerical values within the detailed description and the claims herein are modified by “about” or “approximately” the indicated value, and consider experimental error and variations that would be expected by a person having ordinary skill in the art.

[0109] All documents described herein are incorporated by reference herein, including any priority documents and/or testing procedures to the extent they are not inconsistent with this text. As is apparent from the foregoing general description and the specific embodiments, while forms of the present disclosure have been illustrated and described, various modifications can be made without departing from the spirit and scope of the present disclosure. Accordingly, it is not intended that the present disclosure be limited thereby. Likewise, the term “comprising” is considered synonymous with the term “including.” Likewise whenever a composition, an element or a group of elements is preceded with the transitional phrase “comprising,” it is understood that we also contemplate the same composition or group of elements with transitional phrases “consisting essentially of,” “consisting of,” “selected from the group of consisting of,” or “is” preceding the recitation of the composition, element, or elements and vice versa.

Claims

1. An apparatus for forming a blind via in a substrate, comprising: an optical device comprising: a galvanometer scanner; and a beam expander and collimator; a femtosecond laser beam source configured to direct electromagnetic radiation to the beam expander; a transport assembly configured to position the substrate to receive the electromagnetic radiation; a height sensor configured to detect a height of one or more layers of the substrate; and a controller configured to: receive signals from the height sensor; and control the femtosecond laser beam source based on signals received from the height sensor.
2. The apparatus of claim 1, wherein the galvanometer scanner has a plurality of reflecting facets and an axis of rotation.
3. The apparatus of claim 2, wherein the substrate is positioned to receive the electromagnetic radiation reflected from at least one of the reflecting facets of the galvanometer scanner.
4. The apparatus of claim 1, wherein the femtosecond laser beam source is configured to emit a Gaussian laser beam profile.
5. The apparatus of claim 1, further comprising a positioning sensor configured to detect a leading edge of the substrate.
6. The apparatus of claim 5, wherein the controller is further configured to: receive signals from the positioning sensor; and control the femtosecond laser beam source and the transport assembly based on signals received from the positioning sensor.
7. An apparatus for forming a blind via in a substrate, comprising: an optical device comprising: a galvanometer scanner; and a beam expander and collimator; a femtosecond laser beam source configured to direct electromagnetic radiation to the beam expander; a height sensor configured to detect a height of one or more layers of the substrate; and a controller configured to: receive signals from the height sensor; and control the femtosecond laser beam source based on signals received from the height sensor.
8. The apparatus of claim 7, wherein the galvanometer scanner has a plurality of reflecting facets.
9. The apparatus of claim 8, further comprising a transport assembly configured to position the substrate to receive the electromagnetic radiation.
10. The apparatus of claim 9, wherein the substrate is positioned to receive the electromagnetic radiation reflected from at least one of the reflecting facets of the galvanometer scanner.
11. The apparatus of claim 10, further comprising a positioning sensor configured to detect a

leading edge of the substrate.

12. The apparatus of claim 7, wherein the femtosecond laser beam source is configured to emit a Gaussian laser beam profile.

13. The apparatus of claim 12, wherein the controller is further configured to: receive signals from the positioning sensor; and control the control the femtosecond laser beam source and the transport assembly based on signals received from the positioning sensor.

14. An apparatus for forming a blind via in a substrate, comprising: an optical device comprising: a galvanometer scanner; and a beam expander; a femtosecond laser beam source configured to emit a Gaussian laser beam and to direct electromagnetic radiation to the beam expander; a height sensor configured to detect a height of one or more layers of the substrate; and a controller configured to: receive signals from the height sensor; and control the femtosecond laser beam source based on signals received from the height sensor.

15. The apparatus of claim 14, further comprising a transport assembly configured to position the substrate to receive the electromagnetic radiation.

16. The apparatus of claim 15, wherein the galvanometer scanner has a plurality of reflecting facets and an axis of rotation.

17. The apparatus of claim 16, wherein the substrate is positioned to receive the electromagnetic radiation reflected from at least one of the reflecting facets of the galvanometer scanner.

18. The apparatus of claim 15, further comprising a positioning sensor configured to detect a leading edge of the substrate.

19. The apparatus of claim 18, wherein the controller is further configured to: receive signals from the positioning sensor; and control the control the femtosecond laser beam source and the transport assembly based on signals received from the positioning sensor.

20. The apparatus of claim 14, wherein the optical device further comprises a collimator.
